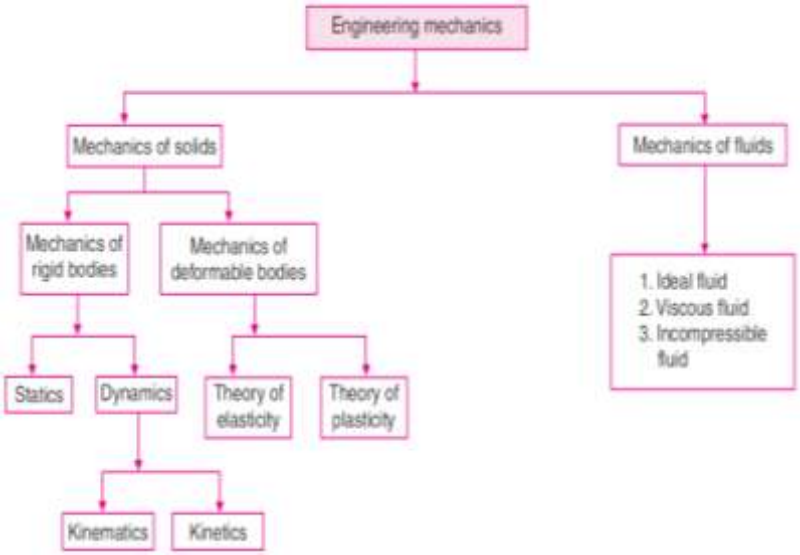
UNIT I

2.INTRODUCTION TO ENGINEERING MECHANICS

2.1 Introduction

Mechanics is a science that describes and predicts the conditions of rest or motion of bodies under the action of forces. Mechanics can be broadly classified as mechanics of solids and mechanics of fluids. Further divisions of mechanics are shown in Fig. 2.1

Fig. 2.1 Branches of Mechanics



Engineering mechanics deals with mechanics of rigid bodies which is further classified as statics and dynamics. **Statics** treats the equilibrium of stationary bodies under the influence of various kinds of forces. **Dynamics**, on the other hand, includes the motion of bodies and forces that causes it.

Study of motion of bodies without considering force causing the motion is called **Kinematics**. On the other hand study of motion of bodies considering the force causing the motion is called **Kinetics**.

If the internal stresses developed in a body are to be studied, the deformation of the body should be considered. This field of mechanics is called **Mechanics of Deformable Bodies/Strength of Materials/Solid Mechanics**. This field may be further divided into **Theory of Elasticity** and **Theory of Plasticity.**

Liquids and gases deform continuously with application of very small shear forces. Such materials are called **Fluids.** The mechanics dealing with behaviour of fluids is called **Fluid Mechanics.**

Mechanics of ideal fluids, mechanics of viscous fluid and mechanics of incompressible fluids are further classification in this area.

2.2 Idealization in Mechanics

A number of ideal conditions are assumed to exist while applying the principles of mechanics to practical problems. In fact without such assumptions it is not possible to arrive at practical solutions. The following idealizations are usually made in engineering mechanics.

a) The body is rigid.

b) The body can be treated as continuum.

c) If the size of the body is small compared to other distances involved in the problem, it may be treated as a particle.

d) If the area over which force is acting on a body is small compared to the size of the body, it may be treated as a point force.

Continuum

A body consists of several matters. It is a well known fact that each particle can be subdivided into molecules, atoms and electrons. It is not possible to solve any engineering problem by treating a body as a conglomeration of such discrete particles. The body is assumed to consist of a continuous distribution of matter. In other words, the body is treated as **continuum.**

Rigid Body

A body is said to be rigid, if the relative positions of any two particles in it do not change under the action of the forces.

Particle

A particle may be defined as an object which has only mass and no size. Such a body cannot exist theoretically. However in dealing with problems involving distances considerably larger compared to the size of the body, the body may be treated as particle, without sacrificing accuracy.

2.3 Basic Principles of Mechanics

The following are the fundamental laws of mechanics:

i. Newton's first law
ii. Newton's second law
iii. Newton's third law

iv. Newton's law of gravitation

v. Law of transmissibility of forces, and

vi. Parallelogram law of forces

2.3.1 Newton's First Law

It states that everybody continues in its state of rest or of uniform motion in a straight line unless it is compelled by external agency acting on it. This leads to the definition of force as the external agency which changes or tends to change the state of rest or uniform linear motion of the body.

2.3.2 Newton's Second Law

It states that the rate of change of momentum of a body is directly proportional to the impressed force and it takes place in the direction of the force acting on it. Thus Second law gives the magnitude of force. ie., F= m . a where, F= resultant force, m= mass of the particle, a = acceleration of the particle.

2.3.3 Newton's Third Law

It states that for every action there is an equal and opposite reaction. Consider the two bodies in contact with each other. Let one body applies a force F on another. According to this law, the second body develops a reactive force R which is equal in magnitude to force F and acts in the line same as F but in the opposite direction.

2.3.4 Newton's Law of Gravitation

The force of attraction between any two bodies is directly proportional to their masses and inversely proportional to the square of the distance between them.

$$F = G\,\frac{m_1 m_2}{d^2}$$

Where 'G' is the constant of proportionality and is known as constant of gravitation.

2.3.5 Law of Transmissibility of Force

According to this law the state of rest or motion of the rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the replaced force (Fig. 2.2) where, P= Force

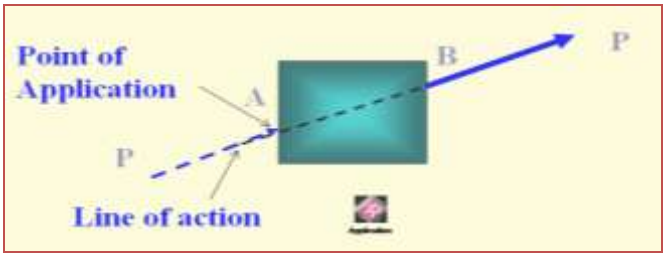


Fig.2.2 Transmissibility of Forces

Principle of Physical Independence of Forces

It states that the action of forces on bodies are independent, in other words the action of forces on a body is not influenced by the action of any other force on the body.

Principle of Superposition of Forces

It states that net effect of a system of forces on a body is same as the combined effect of individual forces acting on the body (Fig. 2.3).

Where, P_1 and P_2 = Forces; a_1 and a_2 = Acceleration due to force

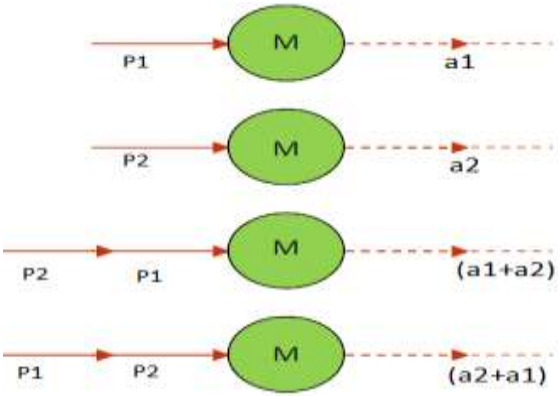


Fig.2.3 Principle of Superposition

2.3.5 Parallelogram Law of Forces

This law states that "if two forces acting simultaneously on a body at a point are represented in magnitude and direction by the two adjacent sides of a parallelogram, their resultant is represented in magnitude and direction by the diagonal of the parallelogram which passes through the point of intersection of the two sides representing the forces".

2.4 Concepts of force

Force: Force is the action exerted by one body upon another. Force tends to change the state of rest or motion of a body to which it is applied. Gravitational force, Push or Pull are the examples of force.

A force is characterized by its points of application, magnitude and direction. A force is represented by vector. Mass is the property of the body which measures the resistance of the body to a change of motion.

2.5 Characteristics of a force

From Newton's first law, we defined the force as the agency which tries to change state of rest or uniform motion of the body. From Newton's second law, we arrived at practical definition of force. Thus 1 Newton is the force required to produce an acceleration of 1 m/sec^2 in a body of 1 kg mass. It may be noted that a force is completely defined only when the following four characteristics are specified

i. Magnitude
ii. Point of application
iii. Line of action, and
iv. Direction

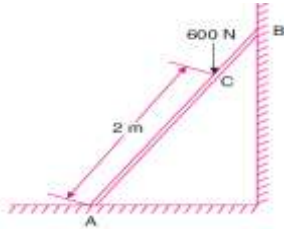


Fig. 2.4 Characteristics of a force

In Fig. 2.4, AB is a ladder leaning against a wall. At point 'C', a person weighing 600 N is standing. The force applied by the person on the ladder has the following characters:

i. Magnitude is 600 N
ii. The point of application is at C which is 2 m from 'A'
iii. The line of action is vertical, and
iv. The direction is downward.

2.6 System of Forces

When number of forces act simultaneously on a body, they are called the system of forces.

Depending upon the orientation of the forces acting on a body, the system of forces is classified as

i) Coplanar force system
ii) Non-coplanar force system.

Coplanar force system consists of a set of forces with their lines of action lying in the same plane. **Non-coplanar force system** consists of a set of forces whose lines of action do not lie in the same plane. Further classifications are indicated in Fig. 2.5 and Fig. 2.6 (a) and Fig. 2.6(b). The types of force system with their characteristics and typical example is listed in Table 2.1(a) and Table 2.1 (b).

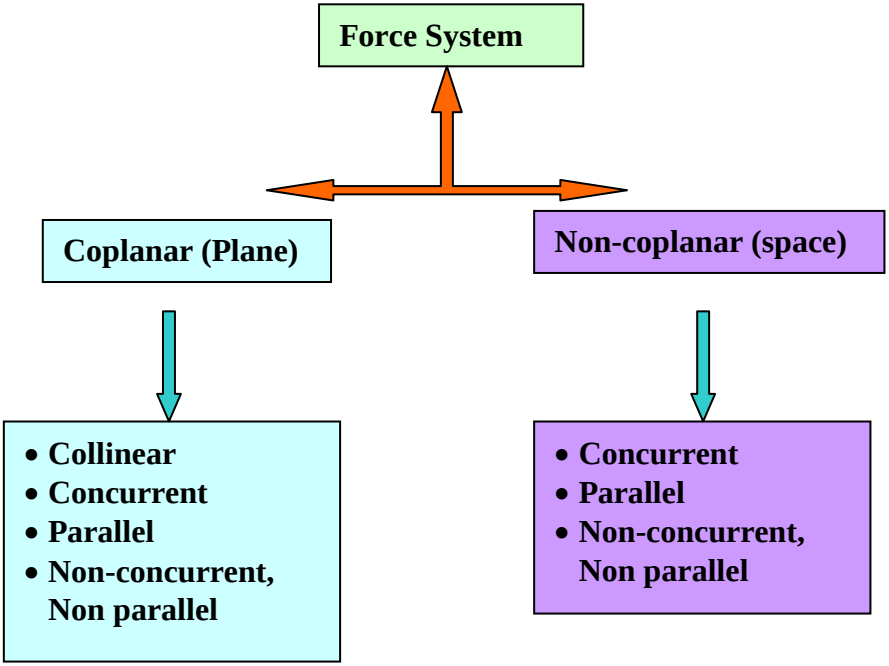


Fig.2.5 Classification of Forces

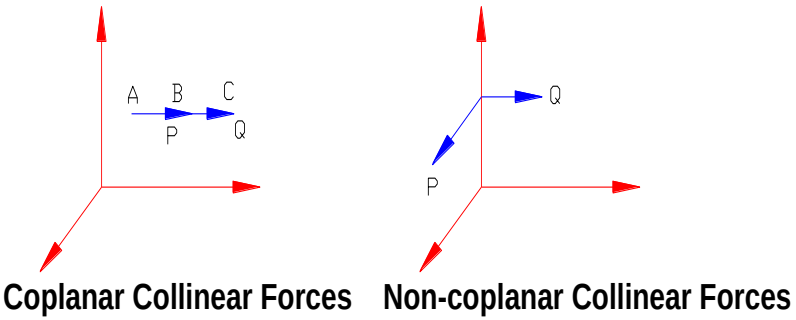


Fig. 2.6 (a) Force Systems

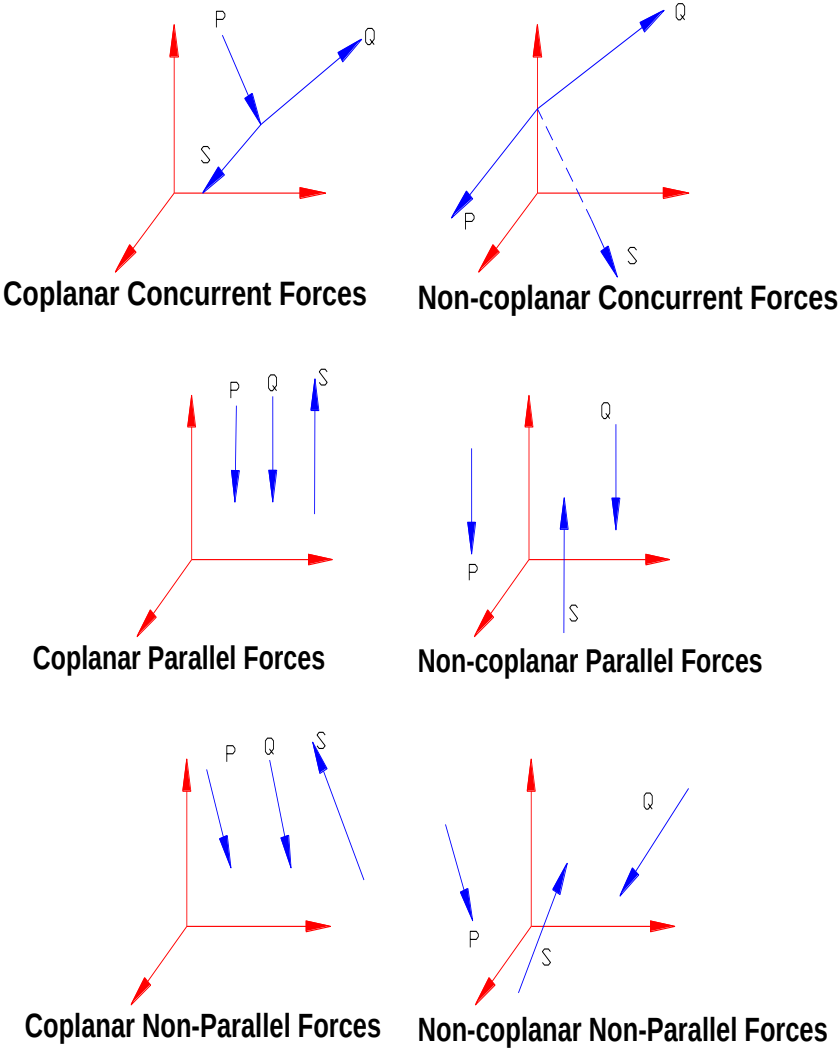


Fig. 2.6 (b) Force Systems

Table 2.1(a) Types of force systems

Force System	Characteristics	Examples
Collinear forces	Line of action of all the forces act along the same line	Forces on a rope in a tug of war
Coplanar Parallel forces	All forces are parallel to each other and lie in a single plane	System of forces acting on a beam subjected to vertical loads
Coplanar like Parallel forces	All forces are parallel to each other and lie in a single plane and act in the same direction	Weight of a stationary train on a rail when the track is straight
Coplanar concurrent forces	Line of action of all forces pass through a single point and forces lie in the same plane	Forces on a rod resting against a wall

Table 2.1(b) Types of force systems

Coplanar non concurrent forces	All forces do not meet at a point but lie in a single plane	Forces on a ladder resting against a wall when a person stands on a rung which is not at its centre of gravity
Non coplanar parallel forces	All the forces are parallel to each other but not in the same plane	The weight of benches in a class room
Non coplanar concurrent forces	All forces do not lie in the same plane ,but their lines of action pass through a single point	A tripod carrying a camera
Non coplanar non concurrent forces	All forces do not lie in the same plane and their lines of action do not pass through a single point	Forces acting on a moving bus

2.7 Moment of a Force

A force applied on a body can create i) a tendency to translate and /or ii) tendency to rotate about an axis or a point. The effect of a force on a body to rotate it about a fixed point is called **Moment**. Moment is measured as the product of force and distance between point of rotation and line of action of force. The point of rotation is called as **Moment Centre** and distance between point of rotation and line of action measured perpendicular to the force is called as **Arm of force** or **Lever Arm**.

For example the force applied by a wrench to a nut can rotate the nut (Fig. 2.7). Similarly the force applied at a door (to open or close) allows it to rotate on its hinge. (Fig. 2.7).



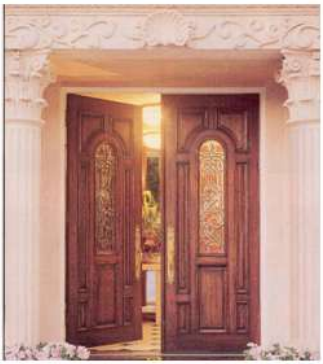


Fig 2.7 Moment of a Force

2.7.1 Moment about a point

The magnitude of moment or tendency of the force to rotate the body about the axis O-O (Fig. 2.8) perpendicular to the plane of the body is proportional to both the magnitude of the force and to the moment arm d, therefore magnitude of the moment is defined as the product of force and moment arm.

Moment = Force x moment arm

M = F×d

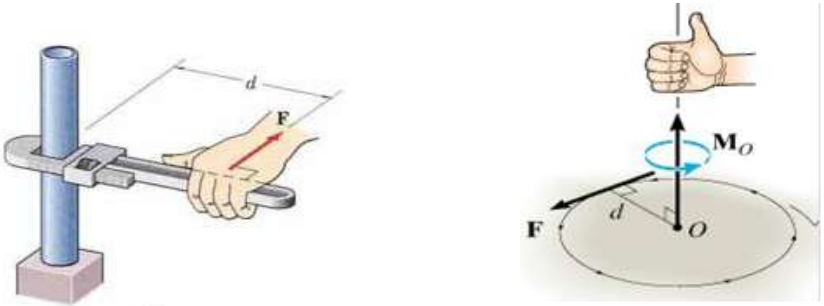


Fig. 2.8 Moment about a point

The moments are classified according to the direction in which the force tends to rotate the body about a fixed point

i. Clockwise and
ii. Anticlockwise moment

Clockwise Moment

When the force tends to rotate the body in the same direction in which the hands of clock move is called clockwise moment.

Anticlockwise Moment

When the force tends to rotate the body in the opposite direction in which the hands of clock move is called anti clockwise moment.

Example 2.1

Find the moment of force F = 600N about A as shown in the Fig. 2.9

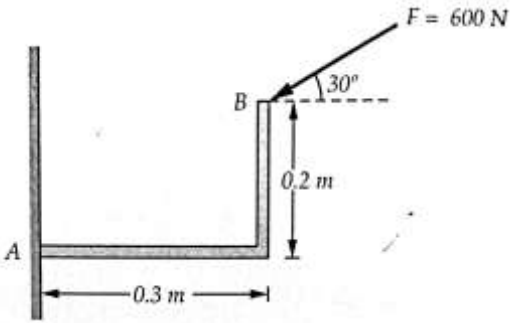


Fig. 2.9

Solution:

Moment of the force can be obtained by resolving the given force as shown in Fig. 2.10

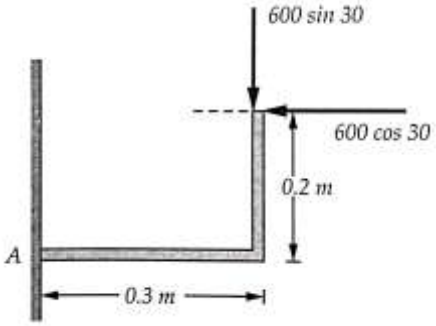


Fig. 2.10

$M_A = (600 \text{ Cos } 30^o)\ (0.2) - (600 \text{ Sin } 30^o)\ (0.3)$

$M_A = 13.92$ N-m (Anticlockwise Moment)

Example 2.2

A 200 N force acts on the bracket as shown in Fig. 2.11 Determine the moment of the force about point A.

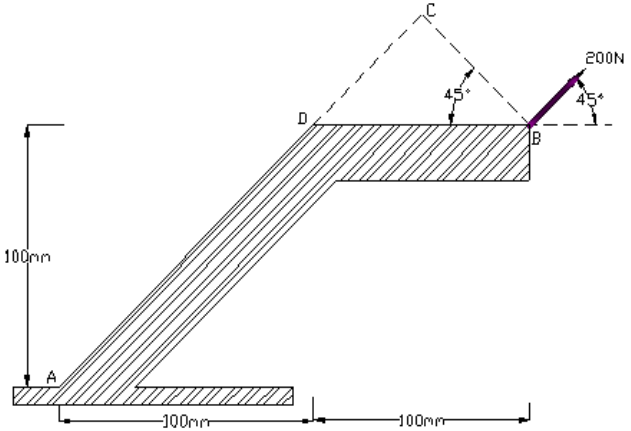


Fig. 2.11

Solution:

Moment about A = M_A= Force x perpendicular distance

=200 x CB (↺)

From triangle BCD CB= DB x cos 45° = 70.71 mm

M_A= 200 x 70.71 = 14142 N-mm = 14.142 N-m

Alternatively:

The force at B is resolved into x and y components and the moment about A is computed using Varignon's theorem

↺ +ve M_A=200 cos 45° x 100 – 200 sin 45° x 200

= -14.142 N-m (↻)

2.8 Couple

Two parallel forces of same magnitude but opposite direction separated by a distance‘d’ as shown in Fig 2.12 form couple.

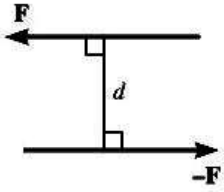


Fig. 2.12 Couple of Force

As matter of fact a couple does not produce any straight-line motion but produces rotation in the body on which it acts. So couple can be defined as unlike parallel forces of same magnitude but opposite direction which produce rotation about a specific direction and whose resultant is zero.

2.8.1 Moment of couple

The moment of the couple is the product of the force (one of the force of the two equal and opposite parallel forces) and the arm of the couple.

Magnitude of the couple is equal to the product of the force and the distance between the parallel forces. M= F x d.

Moment of couple = force x arm of couple

Moment of couple = F x d

Moment of a couple about any point is same.

Let us find the moment of couple about a point ‘O’ on the couple arm AB as shown in Fig 2.13

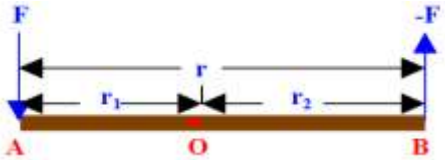


Fig. 2.13 Moment of couple

Moment of Couple about 'O'

$= F \times r_1 + F \times r_2 = F(r_1 + r_2) = F \times r(Anticlockwise)$.

Moment of Couple about 'A'$= F \times r(Anticlockwise)$.

Moment of Couple about 'B'$=. F \times r(Anticlockwise)$

2.9 Resolution of a Force

The process of splitting of a force into its rectangular components (horizontal and vertical) is known as resolution of the force.

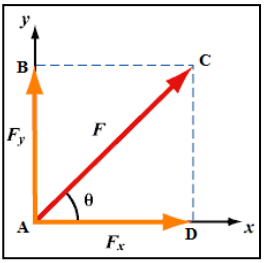


Fig. 2.14 Resolution of a Force

In the Fig. 2.14, F is the force which makes an angle θ with the horizontal axis and has been resolved into two components, namely F_x and F_y,

In ΔCAD,

$$\cos\theta = \frac{F_x}{F} => \boldsymbol{F_x = Fcos\theta}$$

$$\sin\theta = \frac{F_y}{F} => \boldsymbol{F_y = Fsin\theta}$$

2.10 Equivalent force couple system

A force P acting at a point A on a body can be resolved into a force and a couple at any other point B as shown in Fig. 2.15

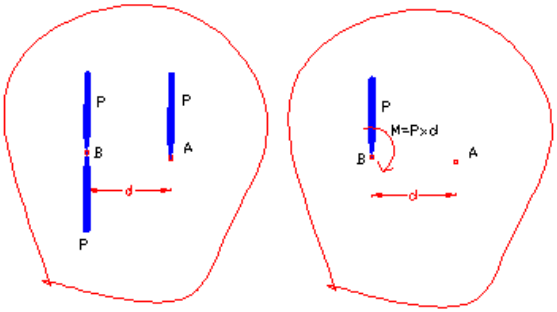


Fig. 2.15 Equivalent force couple system

Equal and opposite force of magnitude P is applied at B and this does not change the equilibrium of the body. The downward force P at A and upward force P at B constitute a couple M = P x at B. Therefore a force P at A is replaces by the same force P and a couple M at B.

Example 2.3

Replace the given system of forces acting on the beam AB shown in the Fig. 2.16 by

a) An equivalent force couple system at A

b) An equivalent force couple system at B.

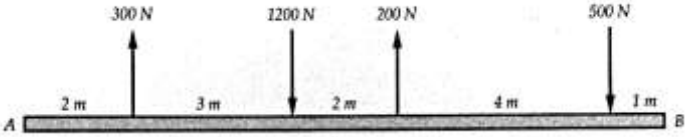


Fig. 2.16

Solution: $R_X = \sum F_X = 0$

$R_y = \sum F_y = 300 - 1200 + 200 - 500$

$R_y = -1200N$

$R = 1200N \downarrow$

$M_A = (300)(2) - (1200)(5) + (200)(7) - (500)(11)$

$M_A = -9500$ Nm

$M_A = 9500$ Nm ↻

The equivalent force couple system at A is as shown in Fig. 2.17



Fig. 2.17

$M_B = (500)(1) - (200)(5) + (1200)(7) - (300)(10)$

$M_B = 4900Nm$ ↺

Example 2.4

Replace the 6 kN force and 10 kN-m moment acting on a beam shown in Fig. 2.18 by a single force. Where does this force cross the x-axis.

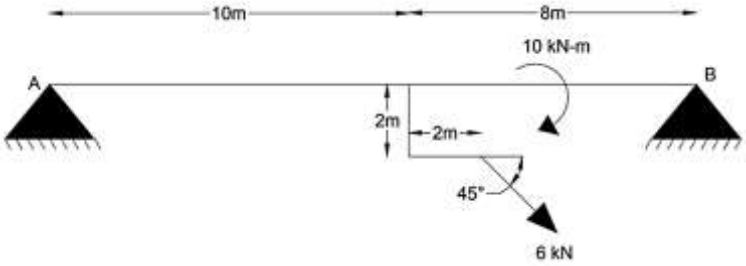


Fig. 2.18

Solution:

Finding the summation of all forces in horizontal and vertical directions

$\sum F_x = 4.24kN \qquad \sum F_y = -4.24$

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2}$$

$R = 6kN \qquad \alpha = 45°$

The summation of moment of all forces about 'A' $\sum M_A = 52.426kN - m$

X-Intercept of the Resultant= $\frac{\sum M_A}{\sum F_y} = 12.35m$ from 'A'

2.11 Composition of Forces

It is the process of combining a number of forces into a single force such that the net effect produced by the single force is equal to the algebraic sum of the effects produced by the individual forces.

The single force in this case is called the resultant force which produces the same effect on the body as that produced by the individual forces acting together.

Sign conventions:

Following sign conventions are used in determining the resultant force

1. **Horizontal forces**:
   i. Left to Right → $^{+ve}$ i. Right to Left $^{-ve}$ ←
2. **Vertical forces**:
   i. Upward ↑ $^{+ve}$ ii. Downward $^{-ve}$ ↓

Example 2.5

Determine the rectangular components of the forces F1 and F2 shown in the Fig. 2.19

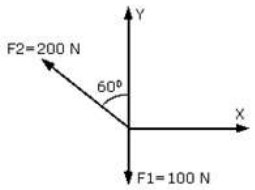


Fig 2.19

Solution:

$F_{1x} = 0$; $F_{1y} = -100N$

$F_{2x} = -200 \cos 30^0$

$F_{2y} = 200 \sin 30^0$

2.12 Resultant of coplanar concurrent force system

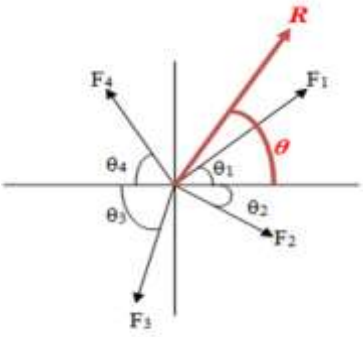


Fig. 2.20

If two or more forces are acting in a single plane and passing through same point such a force system is known as coplanar concurrent force system.

In a coplanar concurrent force system, we can calculate the magnitude and direction of the resultant.

Let, F1, F_2, F_3, F_4 be the concurrent forces acting on a body and making angles θ_1, θ_2, θ_3, θ_4 respectively with the horizontal as shown in the Fig. 2.20.

Steps to find the resultant (Method of Resolution)

i) Resolve all the forces acting on the body into rectangular components.

ii) Calculate the algebraic sum of all the forces acting in the x-direction (horizontal forces).

$$\sum F_x = F_1 \cos\theta_1 + F_2 \cos\theta_2 - F_3 \cos\theta_3 - F_4 \cos\theta_4$$

iii) Calculate the algebraic sum of all the forces acting in the y-direction (vertical forces).

$$\sum F_y = F_1 \sin\theta_1 - F_2 \sin\theta_2 - F_3 \sin\theta_3 + F_4 \sin\theta_4$$

iv) Determine the magnitude of the resultant using the formula

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2}$$

v) Determine the direction of the resultant using the formula

$$\theta = \tan^{-1}\left(\frac{\sum F_y}{\sum F_x}\right)$$

Example 2.6

Determine the magnitude and direction of the resultant of coplanar concurrent force system shown in the figure 2.21.

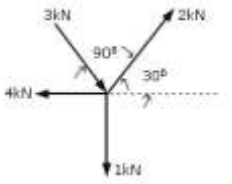


Fig. 2.21

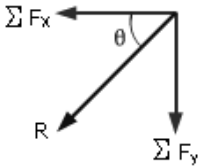
Solution:

$\Sigma Fx = 2 \cos 30^0 - 4 + 3 \cos 60^0 = -0.77kN.$

$\Sigma Fy = 2\sin 30^0 - 1 - 3 \sin 60^0 = -2.59kN.$

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2}$$

$= 2.70kN.$



$$\theta = \tan^{-1}\frac{2.59}{0.77} = 73.44^0$$

Example 2.7

Three forces acting on a hook are as in Fig. 2.22 Determine the direction of the fourth force of magnitude 100N such that the hook is pulled in x direction only. Determine the resultant force in x direction.

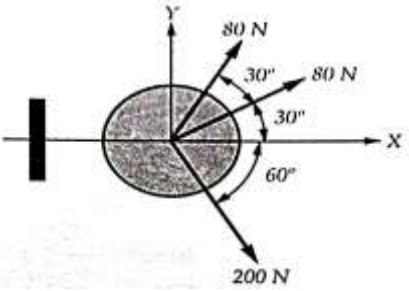


Fig. 2.22

Solution: let the force of 100N act at angle θ with x axis in the first quadrant as shown in the Fig. 2.23

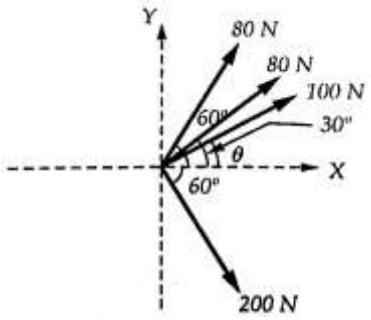


Fig. 2.23

As the resultant is horizontal, its Y-component is zero.

$\sum F_Y = 0$

$100\sin\theta + 80\sin 30 + 80\sin 60 - 200\sin 60 = 0$

$\theta = 39.73°$

$R_X = R = \sum F_X$

$R = 100\cos 39.73 + 80\cos 30 + 80\cos 60 + 200\cos 60$

$R = 286.19kN$

Example 2.8

The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force.

Solution:

The system of given forces is shown in Fig. 2.24

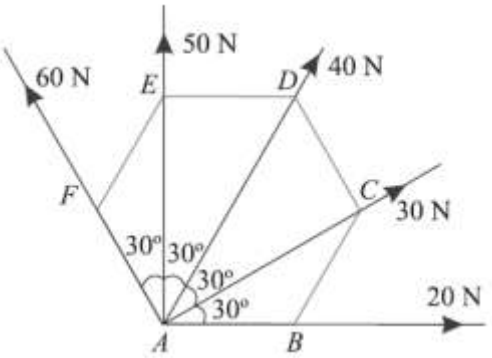


Fig. 2.24

Magnitude of the resultant force

Resolving all the forces horizontally (*i.e.*, along *AB*),

$\Sigma H = 20 \cos 0° + 30 \cos 30° + 40 \cos 60° + 50 \cos 90° + 60 \cos 120°$ N

$= (20 \times 1) + (30 \times 0.866) + (40 \times 0.5) + (50 \times 0) + 60 (-0.5)$ N

$= 36.0$ N

And now resolving the all forces vertically (*i.e.*, at right angles to *AB*),

$\Sigma V = 20 \sin 0° + 30 \sin 30° + 40 \sin 60° + 50 \sin 90° + 60 \sin 120°$ N

$= (20 \times 0) + (30 \times 0.5) + (40 \times 0.866) + (50 \times 1) + (60 \times 0.866)$ N

$= 151.6$ N

We know that magnitude of the resultant force,

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2}$$

= 155.8 N

Direction of the resultant force

Let θ = Angle, which the resultant force makes with the horizontal (*i.e., AB*).

$$\tan\theta = \frac{\Sigma V}{\Sigma H} = \frac{151.6}{36.0} = 4.211$$

θ= 76.6°

Example 2.9

A boat is moved uniformly along a canal by two horses pulling with forces P=890N and Q=1068N acting under an angle α=60° as shown in Fig. 2.25. Determine the magnitude of the resultant pull on the boat and the angles 'β' and 'γ'

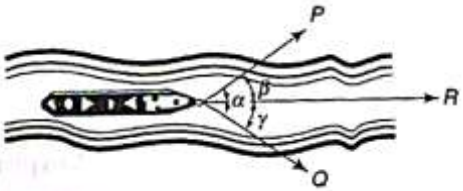


Fig. 2.25

Solution:

Given P=890N and Q=1068N

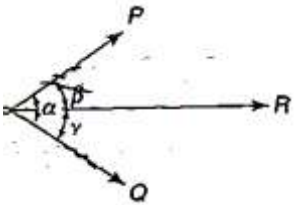


Fig 2.26

Since the Resultant is along the x-Direction, Referring to Fig 2.26 $\sum F_x = R$ and $\sum F_y = 0$

Finding the summation of all the forces along y-Direction and equating $\sum F_y$ to zero, we get

$$P \sin \beta - Q \sin \gamma = 0$$
$$\sin \beta = 1.2 \sin \gamma$$
$$\beta + \gamma = 60°$$
$$\beta = 33°$$
$$\gamma = 27°$$

$$R = \sum F_x = P \cos \beta + Q \cos \gamma$$
$$R = 1698N$$

Example 2.10

A system of forces acting on a body resting on an inclined plane is as shown in Fig. 2.27. Determine the resultant force if θ = 60° and if W = 1000 N; N = 500 N; F = 100 N; and T = 1200 N.

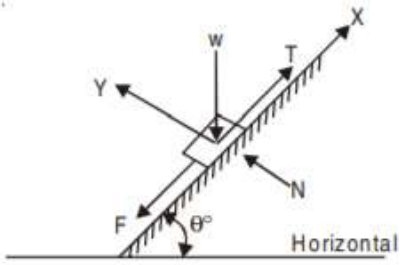


Fig. 2.27

Solution:

In this problem, note that selecting X and Y axes parallel to the plane and perpendicular to the plane is convenient.

$$Rx = \Sigma Fx = T - F - W \sin\theta$$

$$= 1200 - 100 - 1000 \sin 60° = 233.97 \text{ N}$$

$$Ry = \Sigma Fy = N - W \cos 60° = 500 - 1000 \cos 60° = 0.$$

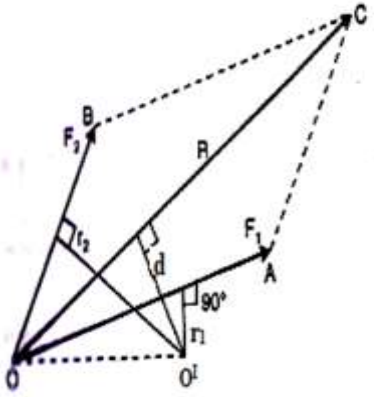
∴ Resultant is force of 233.97 N directed up the plane

2.13 Varignon's Theorem

Varignon's Theorem states that the moment of a force about any point is equal to the algebraic sum of the moments of its components about that point.

Principal of moments states that the moment of the resultant of a number of forces about any point is equal to the algebraic sum of the moments of all the forces of the system about the same point.

Proof:



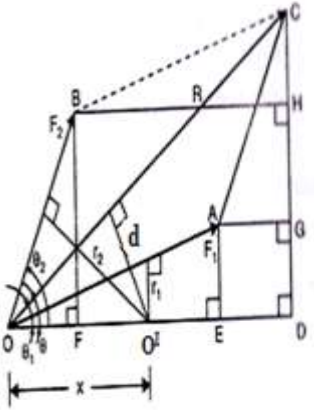


Fig. 2.28(a) Fig 2.28(b)

Fig.2.28 (a) shows two forces F_1 and F_2 acting at point O. These forces are represented in magnitude and direction by OA and OB. Their resultant R is represented in magnitude and direction by OC which is the diagonal of parallelogram OACB. Let 'O' is the point in the plane about which moments of F_1, F_2 and 'R' are to be determined. From point O', draw perpendiculars on OA, OC and OB.

Let r_1= Perpendicular distance between F_1 and O'.

d= Perpendicular distance between R and O'.

r_2= Perpendicular distance between F_2 and O'.

Then according to Varignon's principle,

Moment of R about O' must be equal to algebraic sum of moments of F_1 and F_2 about 'O'.

$R \times d = F_1 \times r_1 + F_2 \times r_2$

Now refer to Fig 2.28(b). Join OO' and produce it to D. From points C, A and B draw perpendiculars on OD to meet at D,E and F respectively. From A and B also draw perpendiculars on CD meeting the line CD at G and H respectively.

Let θ_1 = Angle made by $F_1$with OD,

θ = Angle made by R with OD

θ_2 = Angle made by F_2 with OD.

In Fig 2.28.(b), OA = BC and also OA parallel to BC, hence the projection of OA and BC on the same vertical line CD will be equal i.e., GD = CH as GD is the projection of OA on CD and CH is the projection of BC on CD.

Then, from Fig. 2.28(b), we have

$$P_1 \sin\theta_1 = AE = GD = CH$$

$$F_1 \cos\theta_1 = OE$$

$$F_2 \sin\theta_1 = BF = HD$$

$$F2 \cos\theta_2 = OF = ED$$

(OB = AC and also OB || AC. Hence projections of OB and AC on the same horizontal line OD will be equal i.e., OF = ED)

$$R \sin\theta = CD,\ R \cos\theta = OD$$

Let the length OO' = x.

Then $x \sin\theta_1 = r_1$, $x \sin\theta = d$ and $x \sin\theta_2 = r_2$

Now moment of R about O'

$= R \times$ (distance between O' and R)

$= R \times d$

$= R \times x \sin\theta \quad (d = x \sin\theta)$

$= (R \sin\theta) \times x$

$= CD \times x \quad (R \sin\theta = CD)$

$= (CH + HD) \times x$

$= (F_1 \sin\theta_1 + F2 \sin\theta_2) \times x$

$= F_1 \times x \sin\theta_1 + F_2 \times x \sin\theta_2$

$$= F_1 \times r_1 + F_2 \times r_2$$

$$(x \sin \theta_1 = r_1 \text{ and } x \sin \theta_2 = r_2)$$

= Moment of F_1 about 'O' + Moment of F_2 about 'O'.

Hence moment of R about any point in the algebraic sum of moments of its components F1 and F_2 about the same point. Hence Varignon's principle is proved.

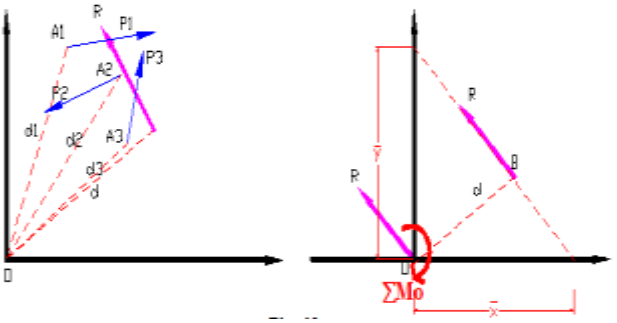
The principle of moments (or Varignon's principle) is not restricted to only two concurrent forces but is also applicable to any coplanar force system, i.e., concurrent or non-concurrent or parallel force system.

2.14 Resultant of coplanar non-concurrent force system

If two or more forces are acting in single plane, but not passing through the single point, such a force system is known as coplanar non-concurrent force system.

Consider a system of forces P_1, P_2, P_3, … etc., acting on a rigid body at the points A_1, A_2, A_3, etc., located at a distance d_1, d_2, d_3 …. etc., as shown in Fig. 2.29 (a)

Each force is replaced by a force of same magnitude, direction and a couple perpendicular to the force at 'O'. Forces P_1, P_2, P_3…etc acting at A1, A2, A3 ….etc., respectively are replaced by respective force and corresponding couple as shown in Fig. 2.29 (b) At 'O' the forces are replaced by the resultant R and net couple ΣM_o. Further this couple ΣM_o and resultant R at O can be reduced to a single force R at the point B such that $\Sigma M_o = R \times d$.



(a) **(b)**

Fig. 2.29

Therefore, $d = \frac{\sum M_o}{R}$, where, $R = \sqrt{R_x^2 + R_y^2}$, $R_x = \Sigma F_x$ and $R_y = \Sigma F_y$. The x and y intercept of the resultant on x and y axes is computed using theorem of moments as $\bar{x} = \frac{M_o}{R_y}$ and $\bar{y} = \frac{M_o}{R_x}$.

In coplanar non-concurrent force system, we can calculate the magnitude, direction and position of the resultant force as follows:

i) Magnitude of resultant using the formula

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2}$$

ii) Direction of the resultant using the formula

$$\theta = \tan^{-1}\left(\frac{\sum F_y}{\sum F_x}\right)$$

iii) Position of resultant

Position of the resultant means the calculation of d, or x and y intercepts as shown in the figure

R x d = algebraic sum of moments of number of forces about that point.

R x d = ΣM_o

$$d = \frac{\Sigma M}{R}$$

X- intercept:

$$x = \left|\frac{\Sigma M}{\Sigma F_y}\right|$$

Y- intercept:

$$y = \left|\frac{\Sigma M}{\Sigma F_x}\right|$$

Example 2.11

Four parallel forces 2 kN, F1, F2 and 10 kN have a downward resultant of 5 kN that acts at a distance of 1.8m to the right of 2kN force as shown in Fig. 2.30. Determine F1 and F2.



Fig. 2.30

Solution

From Fig $\sum F_y = -2 + F_1 - F_2 + 10 = -5$ kN

$F_1 - F_2 = -13$ ---(1)

Summation of Moment of all forces about 'A' is

$\sum M_A = -F_1 + 3F_2 - 40$ ------(2)

Moment of the Resultant about 'A' is $= 5X1.8$ ----(3)

According to varignon's theorem, Moment of the Resultant is equal to the sum of the moments of component forces.

Equating (2) and (3)

$-F_1 + 3F_2 = 49$ -------(4)

Solving (3) and (4)

F_2=18kN F_1=5kN

Example 2.12

Determine the magnitude, direction and position of the resultant force with respect to C as shown in Fig. 2.31.

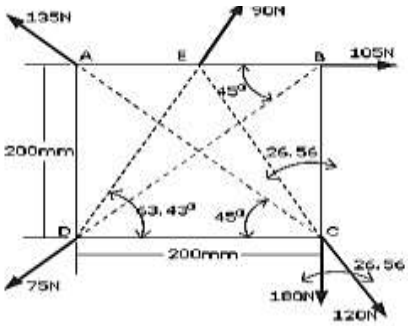


Fig. 2.31

ΣFx = -135 cos 45+ 90 cos 63.43-105+120 cos 63.43-75 cos 45

ΣFx =-159.56 N

ΣFy = 135sin45+90sin 63.43- 120 cos 63.43+180-75sin45

ΣFy =249.25 N

$$R = \sqrt{(-159.65)^2 + (249.25)^2}$$

$$R = 295.95N$$

$$\theta = \tan^{-1}\frac{249.25}{159.65} = 57.37^0$$

ΣM_C = (105x200) - (75 sin 45x200)-(135cos 45x200) + (135 sin 45x200) + (90cos 63.43x 200)+(90sin63.43x100)

ΣM_C = 26494.13 N-mm.

$$d = \frac{\Sigma M}{R} = \frac{26494.13}{295.95} = \mathbf{89.52mm}$$

$$x = \frac{26494.13}{249.25} = \mathbf{106.29mm}$$

$$y = \frac{26494.13}{159.56} = \mathbf{166.04mm}$$

Example 2.13

Determine the resultant of the three forces acting on the dam shown in Fig. 2.32 and locate its intersection with the base AB. For a good design, this intersection should occur within the middle third of the base. Comment whether it is a good design or not.

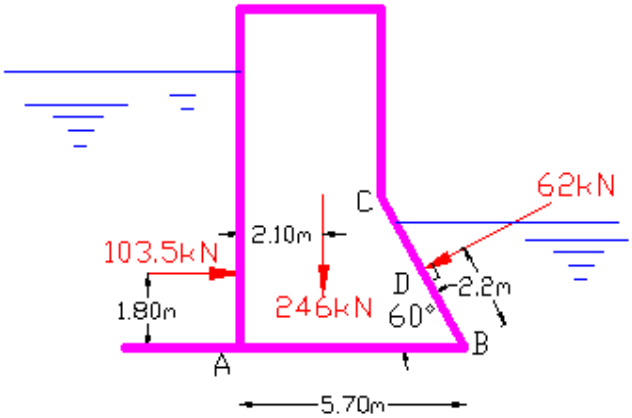


Fig. 2.32

Solution:

$\longrightarrow +^{ve} \Sigma F_x$ 103.5 -62 cos 30=49.8 N

$\uparrow +^{ve} \Sigma F_y =$ -246- 62 sin 30 = -277N

$$R = \sqrt{\sum F_x^{\,2} + \sum F_y^{\,2}} = 281.44\text{N}, \quad \tan\alpha = \frac{\sum F_y}{\sum F_x} = 5.56 \quad \alpha = 79.8^{o}$$

$$\bar{x} = \frac{M_A}{\sum F_y}$$

M_A=103.5 x 1.8 + 246 x 2.1 – 62 x cos30 x 2.2sin60 + 62 sin60 x (5.7 - 2.2 cos60) = 847.5 kN-m

$\therefore \bar{x} = \frac{847.5}{277} = 3.05\text{m}$. For safe design the resultant should act within the middle third. For this the position should be in the range of 1.9m and 3.8 m. Hence the given design is safe.

Example 2.14

Determine the magnitude and Y-intercept of the force system acting on the lamina as shown in Fig. 2.33

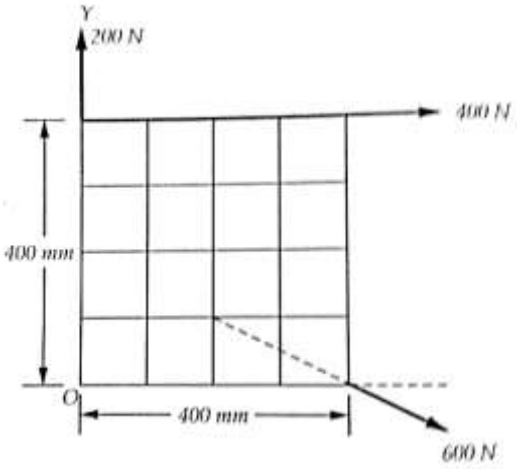


Fig. 2.33

Solution: The angle for the 600N force with horizontal is $\tan^{-1}(\frac{100}{200})$ = 26.565°

$R_X = \sum F_X = 400 + 600\cos 26.565$

$R_X = 936.66N$

$R_y = \sum F_y = 200 - 600\sin 26.565$

$R_y = -68.33N$

$R = \sqrt{R_X^2 + R_Y^2} = \sqrt{936.66^2 + 68.33^2}$

$R = 939.15N$

$\theta = \tan^{-1}(|RY|/|RX|) = \tan^{-1}(68.33/936.66)$

$\theta = 4.17°$

Plot the resultant on Y-axis at distance Y from O as shown in Fig. 2.34 using Varignon's theorem at 'O'

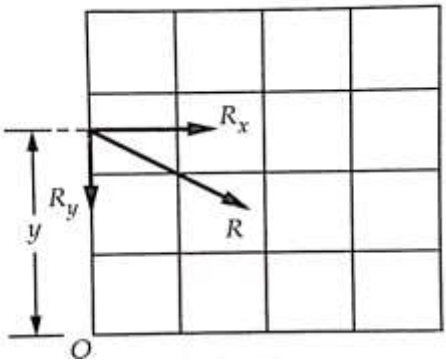


Fig. 2.34

$-(400)(400) - (600\sin 26.565)(400) = -(936.66)(Y)$

$Y = 285.41mm$

Example 2.15

For the non-concurrent coplanar system shown in Fig. 2.35 determine the magnitude, direction and position of resultant force with reference to 'A'.

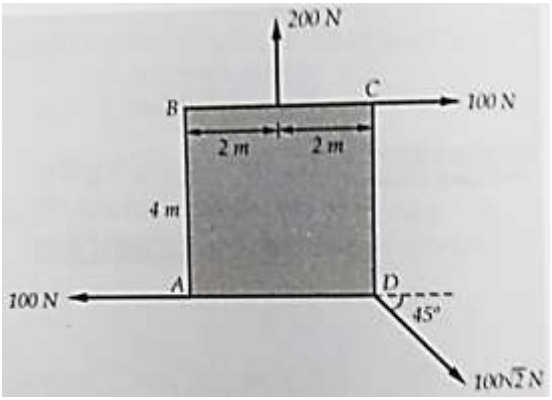


Fig. 2.35

Solution: $R_X = \sum F_X = 100 - 100 + 100\sqrt{2}\cos 45 = 100N$

$R_y = \sum F_y = 200 - 100\sqrt{2}\sin 45 = 100N$

$R = \sqrt{100^2 + 100^2}$

$R = 141.42N$

$\theta = \tan^{-1}(100/100)$

$\theta = 45°$

The perpendicular distance of the resultant from A can be obtained using Varignon's theorem at A

$(200)(2) - (100)(4) - (100\sqrt{2}\sin 45)(4) = (141.42)(d)$

$d = -2.828m$

As the perpendicular distance is negative, the resultant produces clockwise moment about A. The position of resultant with respect to A is shown in Fig. 2.36

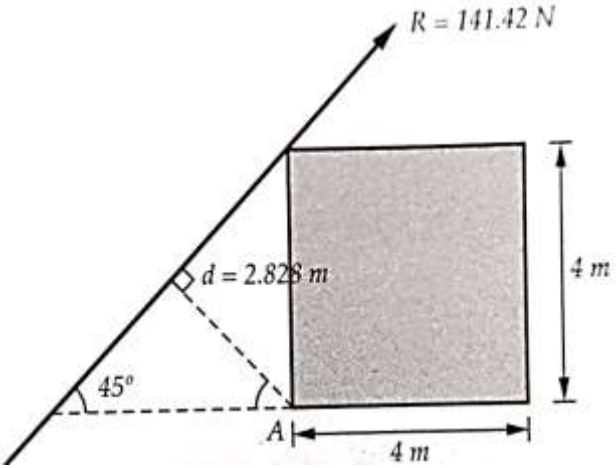


Fig. 2.36

Note the X-intercept will be $\frac{2.828}{sin45}$ = 4m to the left of A and Y-intercept is also 4m above A.

Example 2.16

A rigid plate is subjected to the forces as shown in Fig. 2.37, compute resultant of forces and position of resultant force with respect to centroid point O of the plate.

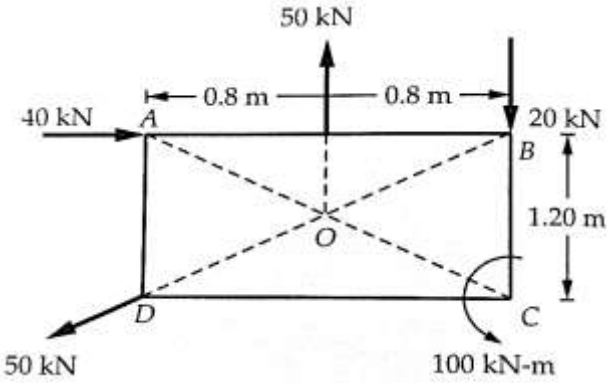


Fig. 2.37

Solution: For the 50kN force at A, the angle with horizontal is $\tan^{-1}(1.2/1.6) = 36.87°$

$R_X = \sum F_X = 40 - 50\cos 36.87 = 0$

$R_y = \sum F_y = 50 - 20 - 50\sin 36.87 = 0$

$R = 0$

As the resultant force is zero, the resultant can be a moment.

$M_O = -(40)(0.6) - (20)(0.8) + 100$

$M_O = 60$kN-m

The resultant is a couple moment of 60kN-m This moment can act anywhere in the plane.

Example 2.17

Determine the magnitude and direction of the resultant for the system of coplanar forces shown in Fig. 2.38. Also determine the position of resultant with respect to 'O'

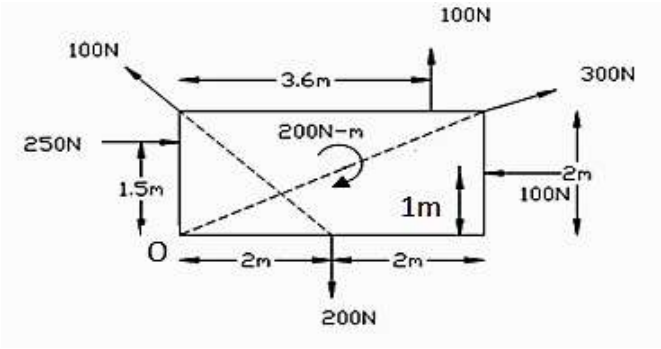


Fig. 2.38

Solution: Referring to the Fig 2.38

$$\sum F_x = 347.62N$$
$$\sum F_y = 104.87N$$
$$R = 363.01N$$
$$\sum M_o = 373.69N - m$$
$$R \times d = \sum M_o$$
$$d = 1.02m$$

Example 2.18 Three forces of 2P, 3P and 4P act along the three sides of an equilateral triangle of side 100 mm taken in order. Find the magnitude and position of the resultant force.

Solution:

The system of given forces is shown in Fig. 2.39

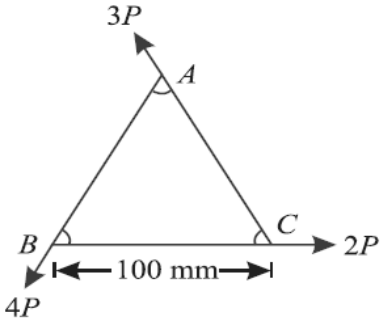


Fig. 2.39

Magnitude of the resultant force

Resolving all the forces horizontally,

$\Sigma H = 2P + 3P \cos 120° + 4P \cos 240°$

$= 2P + 3P (– 0.5) + 4P (– 0.5)$

$= – 1.5 P$

And now resolving all the forces vertically.

$\Sigma V = 3P \sin 60° – 4P \sin 60°$

$= (3P \times 0.866) – (4P \times 0.866)$

$= – 0.866 P$

We know that magnitude of the resultant force

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2} = \sqrt{(-1.5P)^2 + (-0.866P)^2} = 1.732P$$

Position of the resultant force

Let x = Perpendicular distance between B and the line of action of the resultant force.

Now taking moments of the resultant force about B and equating the same,

$1.732\ P \times x = 3P \times 100 \sin 60° = 3P \times (100 \times 0.866) = 259.8\ P$

$$x = \frac{259.8}{1.732} = 150 \text{ mm}$$

2.15 Equilibrium of Forces

A body is said to be under equilibrium, under external forces, when it continues to be at rest or under motion.

2.15.1 Equilibrium of Concurrent force systems:

For a body to be under equilibrium, the resultant of force system should be zero. If a resultant exists, then the resultant of force system should be zero. If a resultant exists, then the body can be under equilibrium by applying a force of equal magnitude and opposite. This force is called as **equilibrant**. For equilibrium $R = \sqrt{\sum F_x^{\ 2} + \sum F_y^{\ 2}} = 0$. To satisfy this condition sum of the components along both directions taken separately should be zero, ie., $\Sigma F_x=0$ and $\Sigma F_y=0$

Equations of equilibrium for coplanar Concurrent force systems:

$\Sigma F_x=0$ and $\Sigma F_y=0$

2.16 Lami's Theorem:

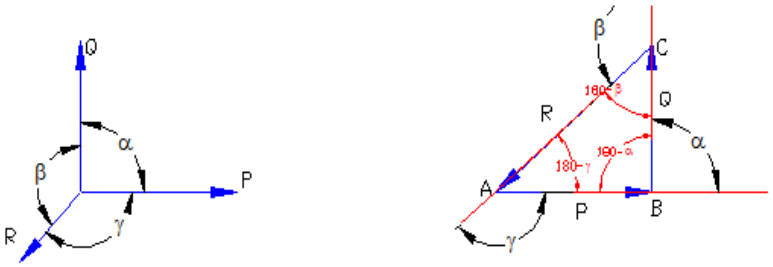


Fig 2.40

Statement: If a body is in equilibrium under action of three forces, each force is proportional to the sine of the angle between the other two forces. Referring to Fig 2.40

$P \propto \sin\beta$; $Q \propto \sin\gamma$; $R \propto \sin\alpha$; or $\dfrac{P}{\sin\beta} = \dfrac{Q}{\sin\gamma} = \dfrac{R}{\sin\alpha}$

Note: In the above equation all the three forces are considered to be of similar nature, ie., either all are acting away (Tensile) or acting towards the point (Compressive). If any of the force is unlike, then that force is considered as negative.

Proof: Applying sine rule to the triangle ABC in Fig. 2.40 the following expression is evaluated

$$\frac{AB}{\sin\angle BCA} = \frac{BC}{\sin\angle CAB} = \frac{CA}{\sin\angle ABC};$$

$$\frac{P}{\sin(180-\beta)} = \frac{Q}{\sin(180-\gamma)} = \frac{R}{\sin(180-\alpha)}$$

$$\frac{P}{\sin\beta} = \frac{Q}{\sin\gamma} = \frac{R}{\sin\alpha}$$

Because sin (180-θ) = sinθ

2.17 Free Body Diagram (FBD)

In practice the problem is derived from actual physical situation. A sketch showing the physical condition of the problem is known as space diagram. In order to obtain solution for this problem, the body under consideration and all the forces acting on the body is considered. Free body diagram is a sketch of isolated body which shows the external forces on the body and the reactions exerted on it by the removed elements. Some examples of writing free body diagram is shown in Fig. 2.41

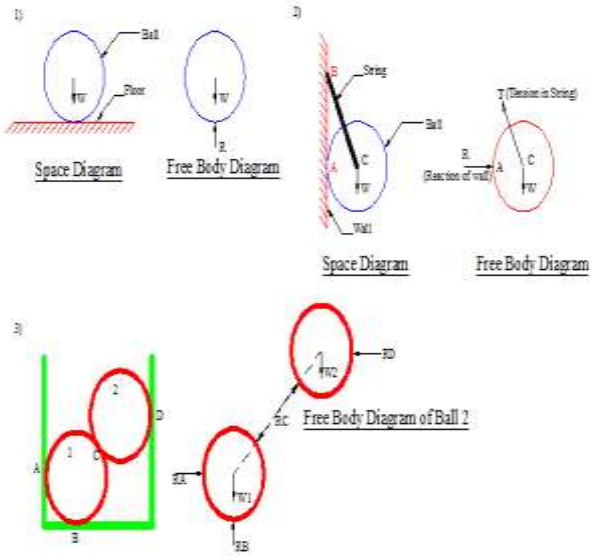


Fig. 2.41

Example 2.19

A sphere of weight 100 N is tied to a smooth wall by a string as shown in Fig. 2.42 Find the tension T in the string and reaction of the wall.

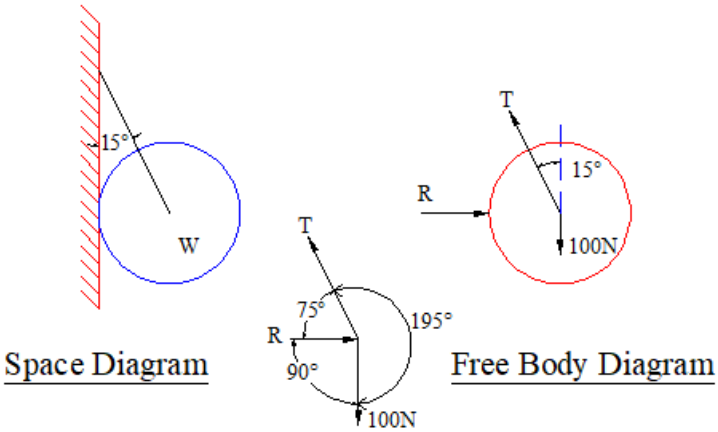


Fig. 2.42

Solution: Free body diagram of the sphere is shown in Fig. 2.42 along with force diagram. Solution is obtained by applying Lami's theorem

$\frac{T}{\sin 90} = \frac{-R}{\sin 195} = \frac{100}{\sin 75}$ Considering two terms at a time the unknowns are computed

$T = \frac{100}{\sin 75} \times \sin 90 = 103.53N$ and $R = \frac{-100}{\sin 75} \times \sin 195 = 26.79N$

Example 2.20

A spherical ball of weight 75 N is attached to a string and is suspended from the ceiling as shown in Fig. 2.43(a) Find the tension in the string, if a horizontal F is applied to the ball as shown in Fig. 2.43(b) Determine the angle the string makes with the vertical and also the tension in the string if F=150 N

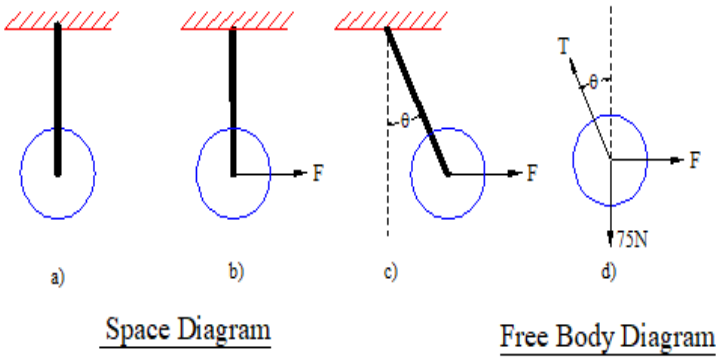


Fig. 2.43(a) Fig. 2.43 (b)

Solution:

i) When the ball is hanging from the ceiling, the tension in the string is equal to the weight of the ball i.e., T=W=75 N.

ii) Free body diagram of ball is shown in Fig. 2.4(b) when a horizontal force F is applied. Magnitude of tensile force and inclination θ of string with vertical can be computed using the method of components and equilibrium condition.

$\longrightarrow +^{ve} \Sigma F_x = 0;\ -T\sin\theta + 150 = 0$ --(1)

$\uparrow +^{ve} \Sigma F_y = 0;\ T\cos\theta - 75 = 0$ --(2)

Equations (1) and (2) can be simplified as $T\sin\theta = 150$; $T\cos\theta = 75$

Dividing first equation by the second equation;

$$\frac{T\sin\theta}{T\cos\theta} = 2;\ \tan\theta = 2 \text{ and } \theta = \tan^{-1} 2 = 63.43^{o}$$

Substituting value of θ in (1) the value of T is computed as T=167.7N

Alternatively, using Lami's theorem, the solution can also be obtained as

$$\frac{T}{\sin 90} = \frac{75}{\sin(90+\theta)} = \frac{150}{\sin(180-\theta)}$$

$T\sin(90+\theta) = 75$ or $T\cos\theta = 75$ and $T\sin(180-\theta) = 150$ or $T\sin\theta = 150$

These two equations are similar to equations (2) and (1) and hence the solution leads to the same result ie $\theta = 63.43°$ and $T = 167.7$ N

Example 2.21

Determine the reactions at contact points for the system shown in Fig. 2.44

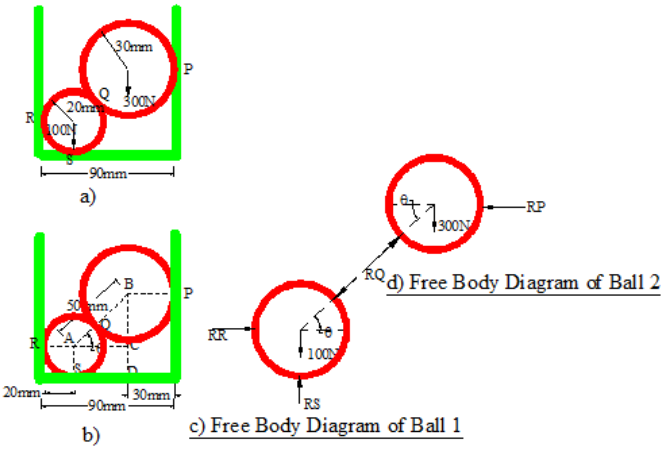


Fig. 2.44

Solution:

The free body diagram of ball 1 and 2 is shown in Fig. 2.44. From triangle ABC the inclination of AB with AC (θ), is computed as $\cos\theta = \frac{AC}{AB}$, where

AC= 90-20-30 = 40mm. and

AB = 50 mm. $\therefore \theta = 36.87°$

Equations of equilibrium is applied to Free Body Diagram of Ball1 as

$\rightarrow +^{ve} \Sigma F_x = 0;\ RR - RQ\cos\theta = 0$ -- (1)

$\uparrow +^{ve} \Sigma F_y = 0;\ RS - 100 - RQ\sin\theta = 0$ --(2)

Equations of equilibrium is also applied to Free Body Diagram of Ball 2 as

$\longrightarrow +^{ve} \Sigma F_x = 0;\ RQ\cos\theta - RP = 0$ --(3)

$\uparrow +^{ve} \Sigma F_y = 0;\ RQ\sin\theta - 300 = 0$ --(4)

From equation (4) $RQ = \dfrac{300}{\sin 36.87} = 500$ substituting this in equation (3), the value of R_D is computed as $R_P = 500 \cos 36.87 = 400$ N

Substituting the value of R_Q in (1) and (2);

$R_R = 500 \cos 36.87 = 400$ N; $\longrightarrow$

$R_S = 100 + 500 \sin 36.87 = 400$ N $\uparrow$

Example 2.22

Two identical cylinders each weighing 2kN are supported by vertical and inclined plane ash shown in Fig. 2.45. Assuming smooth surfaces determine the reactions at A, B and C

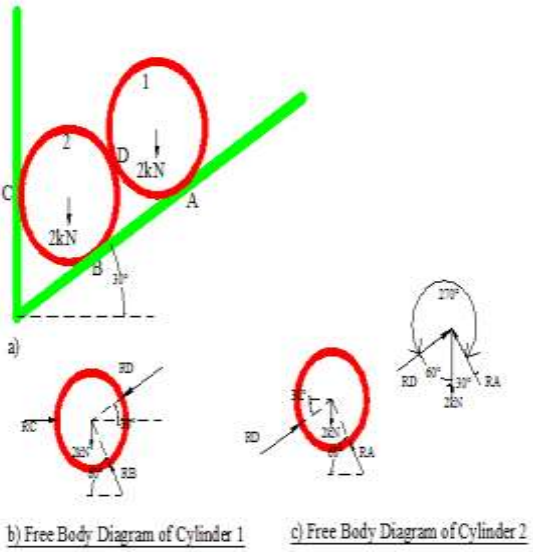


Fig. 2.45

Free body diagram of cylinder 1 and ball 2 is shown in Fig 2.45 (b) and 2.45(c). The solution is obtained by considering first the free body diagram of cylinder 2. Lami's theorem is applied on this as there are three forces in equilibrium

$$\frac{RA}{\sin 60} = \frac{RD}{\sin 30} = \frac{-2}{\sin 270}; \quad R_A = 1.73 \text{ kN}; \; R_D = 1 \text{ kN}$$

Lami's theorem cannot be applied on the free body diagram of cylinder 1, as there are four forces acting on this cylinder. Hence the solution is obtained using the method of components and equilibrium condition.

$\rightarrow +^{ve} \Sigma F_x = 0;\ RC - R_B\cos 60 - R_D\cos 30 = 0$ --(1)

$\uparrow +^{ve} \Sigma F_y = 0;\ -2 + R_B\sin 60 - R_D\sin 30 = 0$ --(2)

Substituting value of RD in (2) , the magnitude of RB is computed as

RB = (2 + 1 sin30)/sin60 = 2.88 kN

Magnitude of RC is computed by substituting RD and RB in(1)

RC= 2.88 cos60+ 1 cos30 =2.31 kN

Example 2.23

A 3 kN is to be supported by the rope and pulley arrangement shown in Fig. 2.46 Determine the magnitude and direction of the force' F', which should be exerted at the free end of the rope.

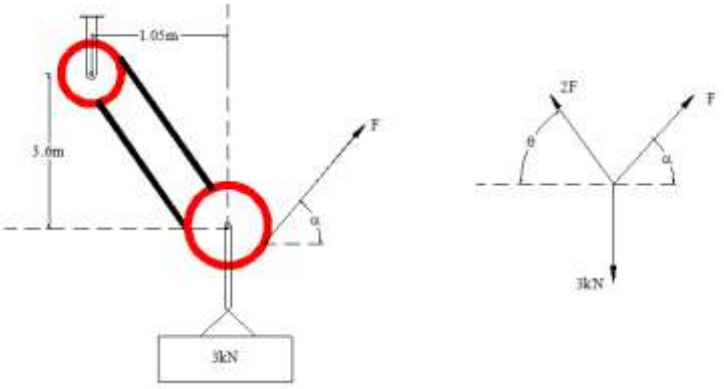


Fig. 2.46

Solution:

In this problem, the size of pulley is ignored. The force F and its inclination α is computed by applying Lami's theorem to the free body diagram shown in Fig 2.46

$$\frac{2F}{\sin(90+\alpha)} = \frac{F}{\sin(90+\theta)} = \frac{3}{\sin(180-(\theta+\alpha))},$$

Here $\theta = \tan^{-1}(\frac{3.6}{1.05}) = 73.74^{o}$

Considering the first two terms, the value of α is computed as

$2F \sin 163.74 = F \cos \alpha \ \therefore \alpha = 55.94°$

Substituting this value in the second and third term:

$$F = \frac{3\sin 163.74}{\sin(180-129.68)}; F = 1.09\,\text{kN}$$

Example 2.24

Five guy wires tied at a point and are pulled in radial directions, equally spaced from one another. If the magnitude of pulls on three consecutive wires is 50 kN, 70 kN and 60 kN determine the magnitude of pulls on two other wires.

Solution:

The forces in guy wires taken in the order is shown in Fig 2.47

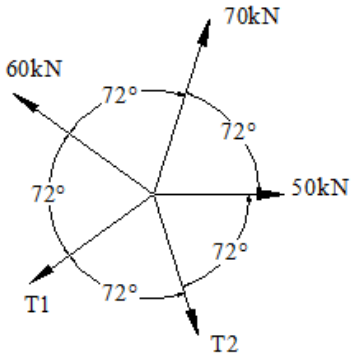


Fig 2.47

The given force system is in equilibrium. The equation of equilibrium is applied to find T1 and T2.

$\longrightarrow$ $+^{ve} \Sigma F_x = 0;$

$50+70\cos72-60\cos36-T_1\cos36+T_2\cos72 = 0$ -- (1)

$\uparrow$ $+^{ve}\Sigma F_y = 0;$

$70\sin72+60\sin36-T_1\sin36-T_2\sin72 = 0$ --(2)

The equations are simplified as

$0.81T_1-0.31T_2 = 23.09$-----(1a)

$0.59T_1+0.95T_2 = 101.84$---(2a)

Solving (1a) and (2a); T1 = 56.18 kN and T2 = 72.31 kN

Example 2.25

A string AB, BC and CD are attached to two fixed point A and D. They have equal weights of 1kN knotted to it at B and C as shown in Fig. 2.48. Find the tension in the portion AB, BC and CD of the strings; also find the inclination of BC with vertical.

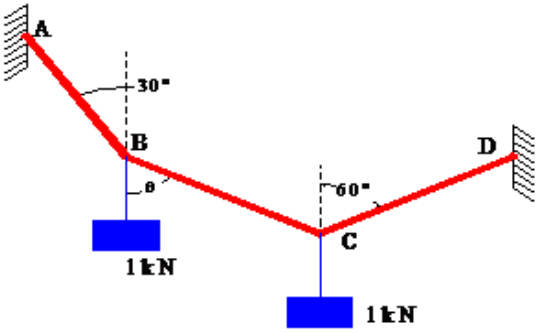


Fig. 2.48

Solution: Let T_{AB}, T_{BC} and T_{CD} be the tensile forces in the string AB, BC and CD respectively.

The forces acting at B and C is shown in Fig. 2.49

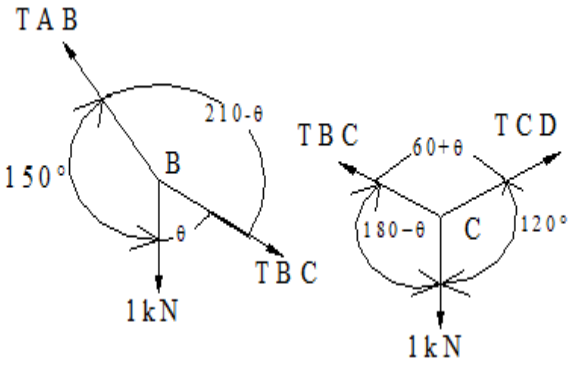


Fig. 2.49

Lami's theorem is applied on the free body diagram at B and C.

$$\frac{TAB}{\sin\theta} = \frac{TBC}{\sin 150} = \frac{1}{\sin(210-\theta)} \quad \text{---(1)}$$

$$\frac{TBC}{\sin 120} = \frac{TCD}{\sin(180-\theta)} = \frac{1}{\sin(60+\theta)} \quad \text{---(2)}$$

Following expressions are obtained after simplifying equation (1)

$$TAB = \frac{\sin\theta}{\sin(210-\theta)} \quad \text{---(3)}$$

$$TBC = \frac{\sin 150}{\sin(210-\theta)} \quad \text{---(4)}$$

$$TBC = \frac{\sin 120}{\sin(60+\theta)} \quad \text{---(5)}$$

$$TCD = \frac{\sin(180-\theta)}{\sin(60+\theta)} \quad \text{---(6)}$$

Value of θ is obtained by equating (4) and (5)

$\therefore$ $\theta = 59.95°$ The force is strings are obtained by substituting this value in (3), (4) and (6)

TAB=1.73 kN; TBC= 1 kN and TCD = 1 kN

Example 2.26

A uniform wheel of 600 mm diameter, weighing 5 kN rests against a rigid rectangular block of 150 mm height as shown in Fig.2.50. Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner A of the block. Also find the reaction on the block. Take all the surfaces to be smooth.

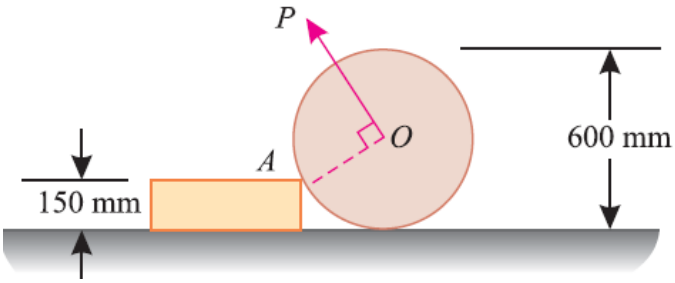


Fig. 2.50

Solution:

Given: Diameter of wheel = 600 mm; Weight of wheel = 5 kN and height of the block = 150 mm.

Least pull required just to turn the wheel over the corner.

Let P = Least pull required just to turn the wheel in kN.

A little consideration will show that for the least pull, it must be applied normal to AO. The system of forces is shown in Fig. 2.51. From the geometry of the figure, we find that

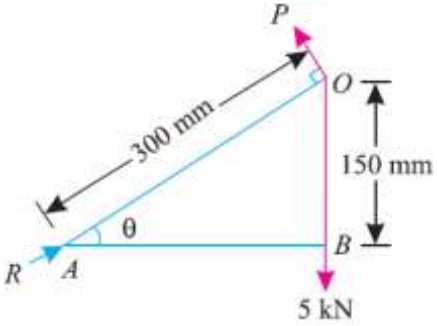


Fig. 2.51

$$\sin\theta = \frac{150}{300} = 0.5$$

$$AB = \sqrt{(300)^2 - (150)^2} = 260 \text{ mm}$$

Now taking moments about *A* and equating the same,

$$P \times 300 = 5 \times 260 = 1300$$

$$P = \frac{1300}{300} = 4.33 \text{ kN}$$

Reaction on the block

Let *R* = Reaction on the block in kN.

Resolving the forces horizontally and equating the same,

$R \cos 30° = P \sin 30°$

$$R = \frac{P \sin 30°}{\cos 30°} = \frac{4.33 \times 0.5}{0.866} = 2.5 \text{ kN}$$

Example 2.27

A light string ABCDE whose extremity A is fixed, has weights W1 and W2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 300 N at the free end E as shown in Fig.2.52. If in the equilibrium position, BC is horizontal and AB and CD make 150° and 120° with BC, find (i) Tensions in the portion AB, BC and CD of the string and (ii) Magnitudes of W1 and W2.

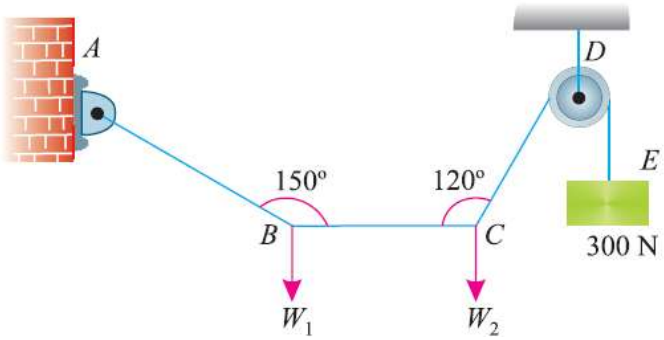


Fig. 2.52

Solution:

Given: Weight at *E* = 300 N

For the sake of convenience, let us split up the string *ABCD* into two parts.

The system of forces at joints *B* and *C* is shown in

Fig. 2.53

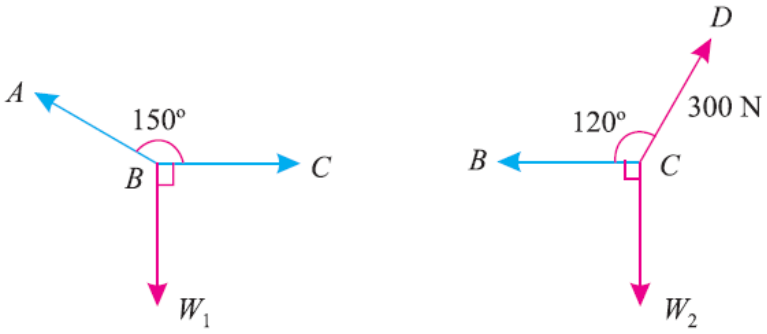


Fig. 2.53

(*i*) *Tensions is the portion AB, BC and CD of the string*

Let T_{AB} = Tension in the portion *AB*, and

T_{BC} = Tension in the portion *BC*,

We know that tension in the portion *CD* of the string.

$T_{CD} = T_{DE} = 300$ N

Applying Lami's equation at C,

$$\frac{T_{BC}}{\sin 150^\circ} = \frac{W_2}{\sin 120^\circ} = \frac{300}{\sin 90^\circ}$$

$$\frac{T_{BC}}{\sin 30^\circ} = \frac{W_2}{\sin 60^\circ} = \frac{300}{1}$$

$T_{BC} = 300 \sin 30^\circ = 300 \times 0.5 = 150$ N

and $W_2 = 300 \sin 60^\circ = 300 \times 0.866 = 259.8$ N

Again applying Lami's equation at *B*,

$$\frac{T_{AB}}{\sin 90^\circ} = \frac{W_1}{\sin 150^\circ} = \frac{T_{BC}}{\sin 120^\circ}$$

$$\frac{T_{AB}}{1} = \frac{W_1}{\sin 30^\circ} = \frac{150}{\sin 60^\circ}$$

$$T_{AB} = \frac{150}{\sin 60^\circ} = \frac{150}{0.866} = 173.2\,\text{N}$$

$$W_1 = \frac{150 \sin 30^\circ}{\sin 60^\circ} = \frac{150 \times 0.5}{0.866} = 86.6\,\text{N}$$

Example 2.28

Three cylinders weighting 100 N each and of 80 mm diameter are placed in a channel of 180 mm width as shown in Fig.2.54. Determine the reactions at all the points of contact.

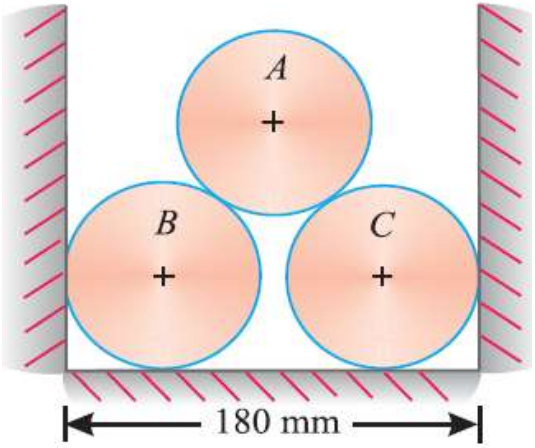


Fig. 2.54

Solution:

Consider the equilibrium of the cylinder *A*. It is in equilibrium under the action of

The following forces, which must pass through the centre of the cylinder as shown in Fig.2.55.

1. Weight of the cylinder 100 N acting downwards.
2. Reaction *R*1 of the cylinder *B* on the cylinder *A*.
3. Reaction *R*2 of the cylinder *C* on the cylinder *A*.

Now join the centres *O*, *P* and *Q* of the three cylinders. Bisect *PQ* at *S* and join *OS* as shown in Fig. 2.55.

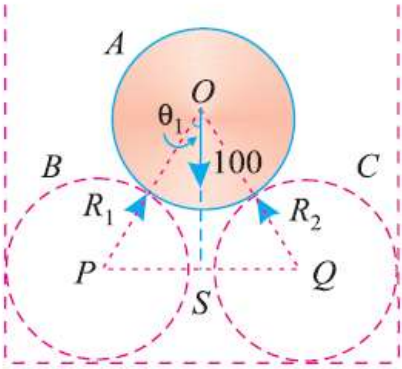


Fig. 2.55

From the geometry of the triangle *OPS*, we find that

OP = 40 + 40 = 80 mm and *PS* = 90 – 40 = 50 mm

$$\sin \angle POS = \frac{PS}{OP} = \frac{50}{80} = 0.625$$

$$\angle POS = 38.7°$$

Since the triangle *OSQ* is similar to the triangle *OPS*, therefore $\angle SOQ$ is also equal to 38.7°. Thus the angle between *R*1 and *R*2 is 2 × 38.7° = 77.4°. And angle between *R*1 and *OS* (also between R_2 and *OS*) = 180° – 38.7° = 141.3° The system of forces at *O* is shown in Fig. 2.56(a). Applying Lami's equation at *O*

$$\frac{R_1}{\sin 141.3^\circ} = \frac{R_2}{\sin 141.3^\circ} = \frac{100}{\sin 77.4^\circ}$$

$$\frac{R_1}{\sin 38.7^\circ} = \frac{R_2}{\sin 38.7^\circ} = \frac{100}{\sin 77.4^\circ}$$

$$R_1 = \frac{100 \times \sin 38.7^\circ}{\sin 77.4^\circ} = \frac{100 \times 0.6252}{0.9759} = 64.0 \text{ N}$$

$R_2 = R_1$

R_3 = Reaction from the wall on the cylinder *B*

R_4 =Reaction from the floor on the cylinder *B*

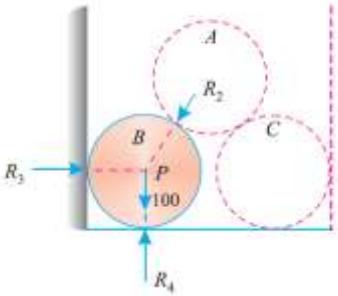


Fig. 2.56 (a)

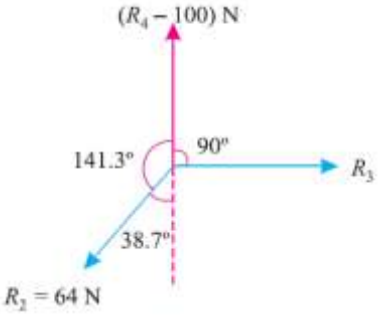


Fig 2.56 (b)

Now consider the equilibrium of the cylinder *B*. It is in equilibrium under the action of the following forces, which must pass through the centre of the cylinder as shown in Fig. 2.56 (*a*).

1. Weight of the cylinder 100 N acting downwards.
2. Reaction *R*2 equal to 64.0 N of the cylinder *A* on the cylinder *B*.
3. Reaction *R*3 of the cylinder *B* on the vertical side of the channel.
4. Reaction *R*4 of the cylinder *B* on the base of the channel.

The system of forces is shown in Fig. 2.56 (*b*). Applying Lami's equation at *P*,

$$\frac{64}{\sin 90^\circ} = \frac{R_3}{\sin (180^\circ - 38.7^\circ)} = \frac{(R_4 - 100)}{\sin (90^\circ + 38.7^\circ)}$$

$$\frac{64}{1} = \frac{R_3}{\sin 38.7^\circ} = \frac{R_4 - 100}{\cos 38.7^\circ}$$

$$R_4 - 100 = 64 \cos 38.7^\circ = 64 \times 0.7804 = 50 \text{ N}$$

$R3 = 64 \sin 38.7^\circ = 64 \times 0.6252 = 40$ N

Example 2.29

The 12 m boom AB weighs 1 kN, the distance of the centre of gravity G being 6 m from A. For the position shown in Fig. 2.57. Determine the tension T in the cable and the reaction at B.

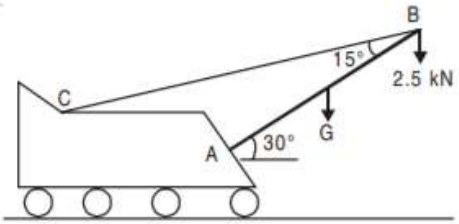


Fig. 2.57

Solution:

The free body diagram of the boom is shown in Fig. 2.58.

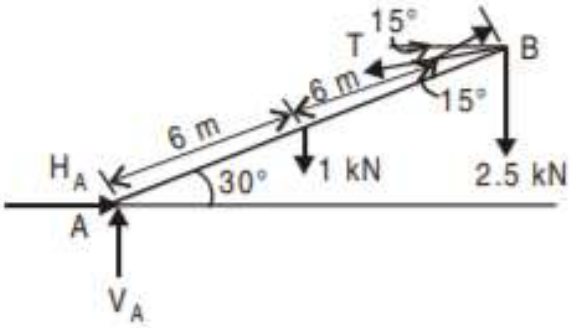


Fig. 2.58

$\Sigma M_A = 0$, gives $T \sin 15° \times 12 - 2.5 \times 12 \cos 30° - 1 \times 6 \cos 30° = 0$

$T = 10.0382$ kN.

$$\sum H = 0, \text{ gives } H_A - T \cos 15° = 0$$

$$H_A = 9.6962 \text{ kN.}$$

$$\sum V = 0, \text{ gives}$$

$$V_A = 1 + 2.5 + T \sin 15° = 6.0981 \text{ kN}$$

$$R_A = \sqrt{V_A^2 + H_A^2}$$

$$R_A = 11.4544 \text{ kN}$$

Example 2.30

A beam 3 m long weighing 400 N is suspended in a horizontal position by two vertical strings, each of which can withstand a maximum tension of 350 N only. How far a body of 200 N weight be placed on the beam, so that one of the strings may just break?

Solution:

The system of given forces is shown in Fig. 2.59

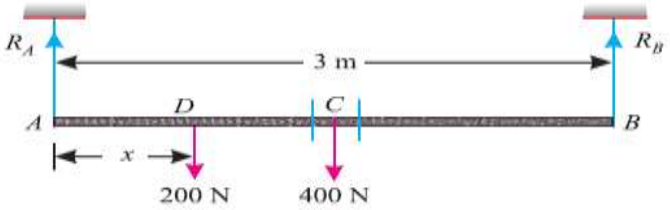


Fig. 2.59

We know that one of the string (say *A*) will just break, when the tension will be 350 N. (*i.e.*, $*R_A$= 350 N). Now taking clockwise and anticlockwise moments about *B* and equating the same,

$350 \times 3 = 200\,(3 - x) + 400 \times 1.5$

$1050 = 600 - 200\,x + 600 = 1200 - 200\,x$

$200\,x = 1200 - 1050 = 150$

$$x = \frac{150}{200} = 0.75\,\text{m}$$

Example 2.31

A block of weight 120N is kept on a smooth inclined plane as shown in Fig.2.60. The plane makes an angle of 32° with horizontal and a force F allied parallel to inclined plane. Compute F and also normal reaction.

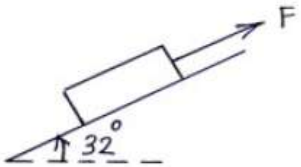
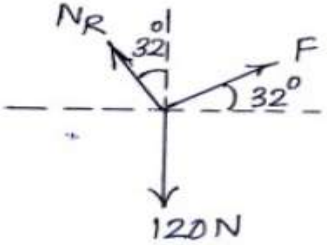


Fig. 2.60

Solution:

Draw FBD



Applying Lami's theorem to the system of forces

$$\frac{120}{Sin90^o} = \frac{F}{Sin(180-32)^o} = \frac{NR}{Sin(90+32)^o}$$

$$F = 63.59N$$

$$N_R = 101.76N$$

Example 2.32

Determine the horizontal force P to be applied to a block of weight 1500 N to hold, it in position on a smooth inclined plane AB which makes an angle of 30° with the horizontal Fig. 2.61.

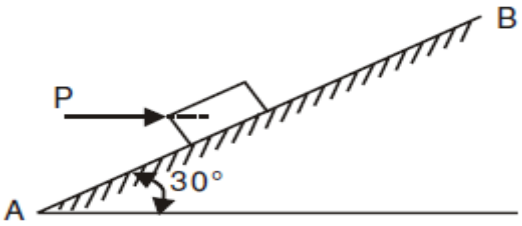
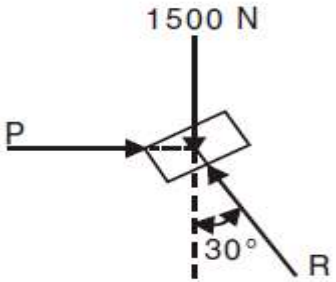


Fig. 2.61

Solution:

Draw FBD



Applying Lami's theorem to the system of forces

$$\frac{R}{\sin 90^\circ} = \frac{P}{\sin(180-30)} = \frac{1500}{\sin(90+30)}$$

$$\boldsymbol{R = 1732.06 \text{ and } P = 866.03.}$$

The above problem may be solved using equations of equilibrium also

$$\Sigma F_y = 0, \text{ gives}$$
$$R \cos 30^\circ - 1500 = 0$$
$$\boldsymbol{R = 1732.06 \text{ N.}}$$
$$\Sigma F_x = 0, \text{ gives}$$
$$P - R \sin 30^\circ = 0$$
$$P = R \sin 30^\circ$$
$$\boldsymbol{P = 866.03 \text{ N.}}$$

Example 2.33

A system of connected flexible cables shown in Fig. 2.62 is supporting two vertical forces 200 N and 250 N at points B and D. Determine the forces in various segments of the cable.

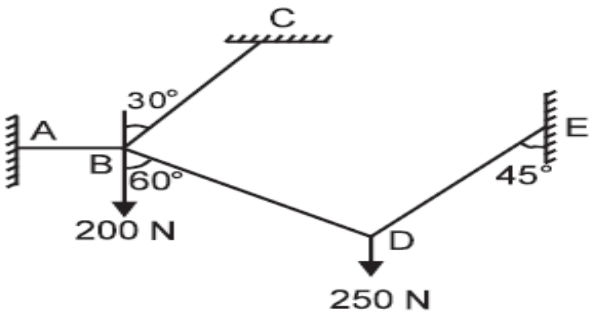
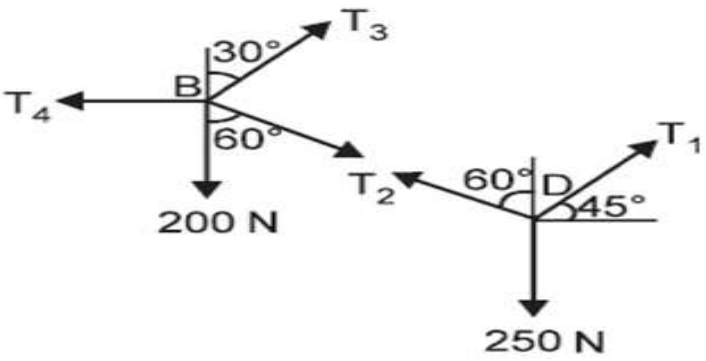


Fig. 2.62

Solution:

Draw FBD



Applying Lami's theorem to the system of forces at point *D*,

$$\frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 135^\circ} = \frac{250}{\sin 105^\circ}$$

$\therefore$ $\quad T_1 = \mathbf{224.14\ N.}$

$\quad T_2 = \mathbf{183.01\ N.}$

Consider the system of forces acting at *B*.

$$\Sigma V = 0$$

$$T_3 \cos 30^\circ - 200 - T_2 \cos 60^\circ = 0$$

$$T_3 = \frac{200 + 183.01 \cos 60^\circ}{\cos 30^\circ}$$

$$T_3 = \mathbf{336.60\ N.}$$

$$\Sigma H = 0$$

$$T_4 - T_2 \sin 60^\circ - T_3 \sin 30^\circ = 0$$

$$T_4 = 183.01 \times \sin 60^\circ + 336.60 \sin 30^\circ$$

$$T_4 = \mathbf{326.79\ N.}$$

Example 2.34

Two cylinders, A of weight 4000 N and B of weight 2000 N rest on smooth inclines as shown in Fig. 2.63 They are connected by a bar of negligible weight hinged to each cylinder at its geometric centre by smooth pins. Find the force P to be applied as shown in the figure such that it will hold the system in the given position.

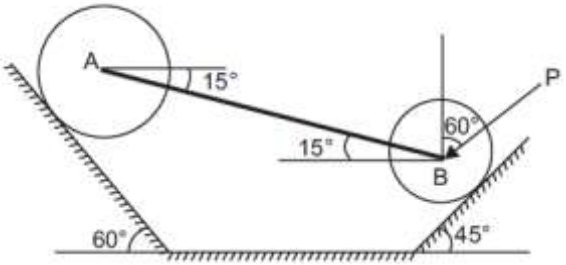
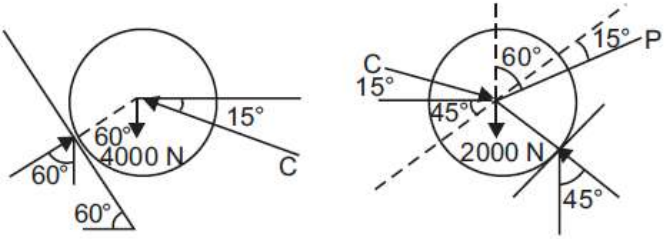


Fig. 2.63

Solution:

Draw FBD



Applying Lami's theorem to the system of forces on cylinder *A*, we get

$$\frac{C}{\sin 60^\circ} = \frac{4000}{\sin(60+90-15)}$$

$$C = 4898.98 \text{ N}$$

Consider cylinder *B.*

Summation of the forces parallel to the inclined plane (45° to horizontal) = 0, gives:

$$P \cos 15° + 2000 \cos 45° - C \cos 60° = 0$$

$$P = \frac{4898.98 \cos 60° - 2000 \cos 45°}{\cos 15°}$$

$$P = 1071.8 \text{ N.}$$

Example 2.35

A cable car used for carrying materials in a hydroelectric project is at rest on a track formed at an angle of 30° with the vertical. The gross weight of the car and its load is 60 kN and its centroid is at a point 800 mm from the track half way between the axles. The car is held by a cable as shown in Fig. 2.64 The axles of the car are at a distance 1.2 m. Find the tension in the cables and reaction at each of the axles neglecting friction of the track.

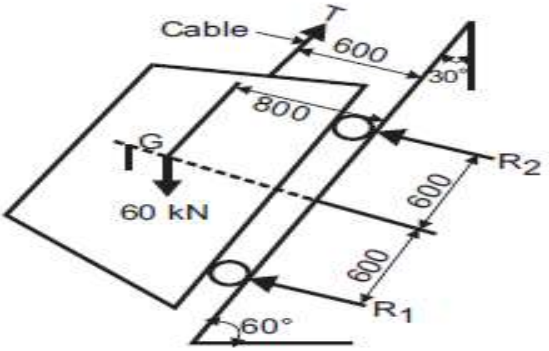


Fig. 2.64

Solution:

Let T be the tension in the cable and the reaction at the pair of wheels be R_1 and R_2 as shown in Fig. 2.64

Now, Σ of forces parallel to the track = 0, gives

$$T - 60 \sin 60^\circ = 0$$

$$\boldsymbol{T = 51.9615 \text{ kN.}}$$

Taking moment equilibrium condition about upper axle point on track, we get

$$R_1 \times 1200 + T \times 600 - 60 \sin 60^\circ \times 800 - 60 \cos 60^\circ \times 600 = 0$$

$$\boldsymbol{R_1 = 23.6603 \text{ kN.}}$$

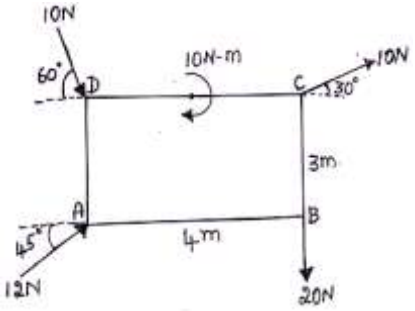
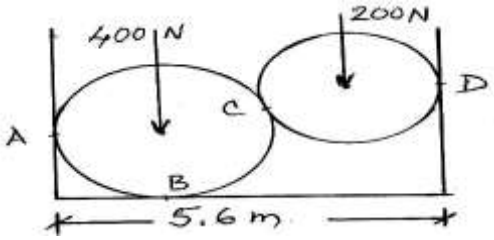
Σ of forces normal to the plane = 0, gives

$$R_1 + R_2 - 60 \cos 60^\circ = 0$$

$$R_2 = 30 - 23.6603$$

$$\boldsymbol{R_2 = 6.3397 \text{ kN.}}$$

Q. No.	Objective Questions
1	The resultant force of two concurrent forces becomes maximum and minimum if angle between them is_____
2	Differentiate between Resultant force and Equilibriant
3	A block weighing 10KN rests on an inclined plane of inclination 30^0 with the horizontal. The magnitude of the component of weight parallel to inclined plane is _____
4	Define Free body diagram
5	State Conditions of equilibrium for coplanar non concurrent force system
6	A 500 N force makes an angle of 30^0 with the Y-axis in second quadrant. Its x-component is _____.
7	The resultant of two forces each equal to P acting at right angles is _____.
8	The process of finding out resultant force is called _____.
9	If the relative position of two points in a body does not change after the application of external force, the body is said to be _____.
10	Principle of transmissibility of forces is applicable for _____.
11	State and prove Lami's Theorem
12	Lamis theorem can be applied when the maximum number of unknowns is _____.
13	The equations of equilibrium of coplanar non- concurrent force system are _____.
14	What is system of forces? Define Coplanar force system and concurrent non planar force system.
15	State the conditions of static equilibrium.
16	Define force. Mention the elements of force.

Q. No.	Descriptive Questions
1	Determine magnitude, direction and position of the resultant for the force system acting on the plate as shown in **Fig. 1**. with respect to 'A'.  **Fig. 1**
2	Two cylinders of radii 2m and 1m are placed in a rectangular box as shown in **Fig 2.**  **Fig. 2** Neglecting the effect of friction, determine the reactions at contact surfaces.
3	A string is subjected to the force of 6KN and 'P' as shown in **Fig.3**. Determine the magnitude of 'P' and the tensions induced in the various portions of the string using Lamis theorem.

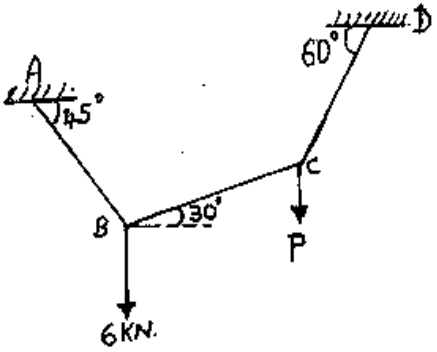


Fig. 3

4 Two spheres each of radius 100mm and weight 5 KN is in a rectangular box as shown in **Fig.4**. Calculate the reactions at all the points of contact.

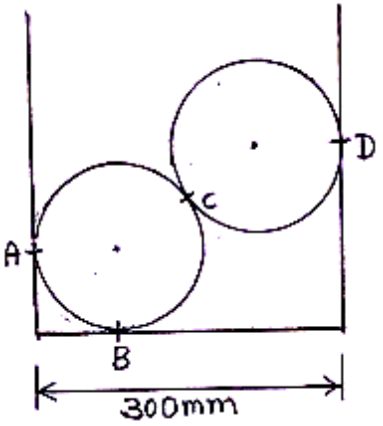
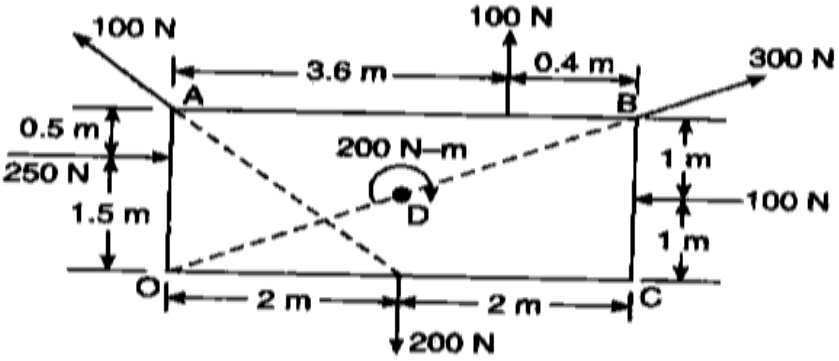
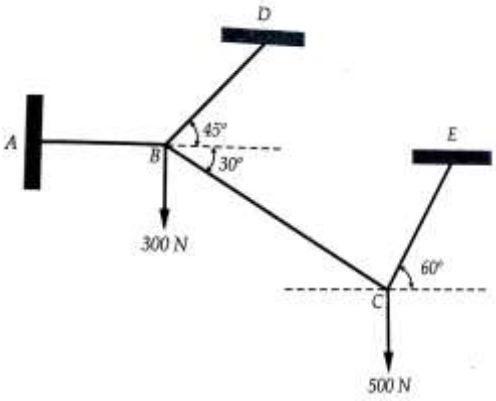
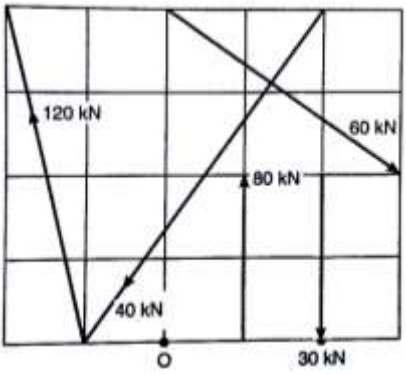
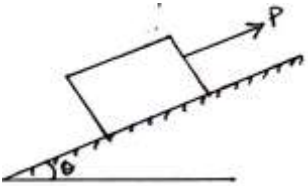
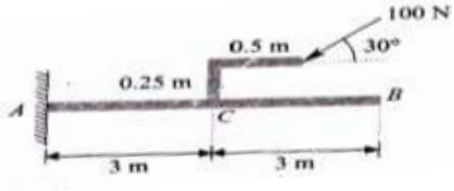
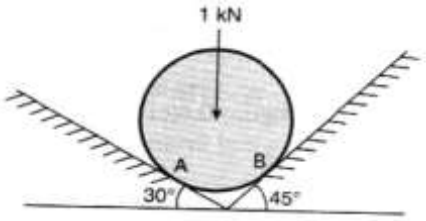


Fig. 4

5	Determine magnitude, direction and position of the resultant for the force system acting on the plate as shown in **Fig. 5**. with respect to 'A'.  Fig. 5
6	A system of cable is in equilibrium condition under two vertical loads 300 N and 500 N as shown in Fig.6. Determine the forces developed in the various portions of the string.  Fig. 6
7	Determine the magnitude direction and position of the resultant force

	for system of forces shown in Fig.7 with respect to point O.  Fig. 7
8	A block of Weight 200N, resting on an inclined plane is in equilibrium by applying a force 'P' as shown in Fig.8 Draw the free body diagram for the block.  Fig. 8
9	Determine the Moment of 100 N force about A, shown in Fig.9  Fig. 9

10	Draw the Free Body Diagram of sphere shown in Fig. 10 and determine the reactions at the points of contact using Lamis theorem  Fig. 10